



| IN                   | anat_template_wf | OUT                         |
|----------------------|------------------|-----------------------------|
|                      |                  | outputnode.anat_realign_xfm |
| inputnode.anat_files |                  | outputnode.anat_ref         |
|                      |                  | outputnode.anat_valid_list  |
|                      |                  | outputnode.out_report       |

| inputnode    | OUT |
|--------------|-----|
| flair        |     |
| roi          |     |
| subject_id   |     |
| subjects_dir |     |
| t1w          |     |
| t2w          |     |

| IN           | fs_jsrunning | OUT |
|--------------|--------------|-----|
| subject_id   | utility      |     |
| subjects_dir |              | out |

| IN      | anat_validate | OUT      |
|---------|---------------|----------|
| in_file | header        | out_file |

| IN                 | brain_extraction_wf | OUT                       |
|--------------------|---------------------|---------------------------|
|                    |                     | outputnode.bias_corrected |
| inputnode.in_files |                     | outputnode.out_file       |
|                    |                     | outputnode.out_mask       |
|                    |                     | outputnode.out_segm       |

| IN          | t1w_buffer | OUT         |
|-------------|------------|-------------|
| ants_seg    |            | ants_seg    |
| t1w_brain   | utility    | t1w_brain   |
| t1w_mask    |            | t1w_brain   |
| t1w_preproc |            | t1w_preproc |

| IN           | sourcefile_buffer | OUT          |
|--------------|-------------------|--------------|
|              | utility           |              |
| source_files |                   | source_files |

| IN                   | t2w_template_wf | OUT                 |
|----------------------|-----------------|---------------------|
| inputnode.anat_files |                 | outputnode.anat_ref |

| IN                         | surface_recon_wf | OUT                         |
|----------------------------|------------------|-----------------------------|
| inputnode.flair            |                  | outputnode.fsnative2t1w_xfm |
| inputnode.skullstripped_t1 |                  |                             |
| inputnode.subject_id       |                  | outputnode.subject_id       |
| inputnode.subjects_dir     |                  | outputnode.subjects_dir     |
| inputnode.t1w              |                  |                             |
| inputnode.t2w              |                  |                             |

| IN                      | ds_template_wf | OUT                     |
|-------------------------|----------------|-------------------------|
| inputnode.anat_preproc  |                | outputnode.anat_preproc |
| inputnode.anat_ref_xfms |                |                         |
| inputnode.source_files  |                |                         |

| IN                          | ds_fs_registration_wf | OUT                          |
|-----------------------------|-----------------------|------------------------------|
| inputnode.fsnative2anat_xfm |                       | outputnode.fsnative2anat_xfm |
| inputnode.source_files      |                       |                              |

| IN           | bbreg | OUT          |
|--------------|-------|--------------|
| source_file  |       | out_ita_file |
| subject_id   |       |              |
| subjects_dir |       |              |

| IN                     | gifti_spheres_wf | OUT                   |
|------------------------|------------------|-----------------------|
| inputnode.subject_id   |                  | outputnode.sphere     |
| inputnode.subjects_dir |                  | outputnode.sphere_reg |

| IN                     | gifti_morphometrics_wf | OUT                  |
|------------------------|------------------------|----------------------|
| inputnode.subject_id   |                        | outputnode.sulc      |
| inputnode.subjects_dir |                        | outputnode.thickness |

| IN                     | ds_spheres_wf | OUT                   |
|------------------------|---------------|-----------------------|
| inputnode.source_files |               | outputnode.sphere     |
| inputnode.sphere       |               | outputnode.sphere_reg |
| inputnode.sphere_reg   |               |                       |

| IN                     | ds_morph_wf | OUT                  |
|------------------------|-------------|----------------------|
| inputnode.source_files |             | outputnode.sulc      |
| inputnode.sulc         |             |                      |
| inputnode.thickness    |             | outputnode.thickness |

| IN                | fs2anat_buffer | OUT               |
|-------------------|----------------|-------------------|
| fsnative2anat_xfm | utility        | fsnative2anat_xfm |

| IN                          | refinement_wf | OUT                      |
|-----------------------------|---------------|--------------------------|
| inputnode.ants_segs         |               | outputnode.out_brainmask |
| inputnode.fsnative2anat_xfm |               |                          |
| inputnode.reference_image   |               |                          |
| inputnode.subject_id        |               |                          |
| inputnode.subjects_dir      |               |                          |

| IN  | merge_xfms | OUT |
|-----|------------|-----|
| in1 | utility    | out |
| in2 |            |     |

| IN                          | gifti_surface_wf | OUT                     |
|-----------------------------|------------------|-------------------------|
| inputnode.fsnative2anat_xfm |                  | outputnode.midthickness |
| inputnode.subject_id        |                  | outputnode.pial         |
| inputnode.subjects_dir      |                  | outputnode.white        |

| IN                     | ds_surfaces_wf | OUT                     |
|------------------------|----------------|-------------------------|
| inputnode.midthickness |                | outputnode.midthickness |
| inputnode.pial         |                | outputnode.pial         |
| inputnode.source_files |                |                         |
| inputnode.white        |                | outputnode.white        |

| IN           | surfaces_buffer | OUT          |
|--------------|-----------------|--------------|
| midthickness |                 | midthickness |
| pial         |                 | pial         |
| sphere       | utility         | sphere       |
| sphere_reg   |                 | sphere_reg   |
| sulc         |                 | sulc         |
| thickness    |                 | thickness    |
| white        |                 | white        |

| IN        | applyrefined | OUT      |
|-----------|--------------|----------|
| in_file   | fsl          | out_file |
| mask_file |              |          |

| IN      | t2wtot1w_xfm | OUT     |
|---------|--------------|---------|
| in_xfms | nitransforms | out_xfm |

| IN              | t2w_resample | OUT          |
|-----------------|--------------|--------------|
| input_image     | fixes        | output_image |
| reference_image |              |              |
| transforms      |              |              |

| IN        | refined_buffer | OUT       |
|-----------|----------------|-----------|
| t1w_brain | utility        | t1w_brain |
| t1w_mask  |                | t1w_mask  |

| IN          | ds_t2w_preproc | OUT      |
|-------------|----------------|----------|
| in_file     |                | out_file |
| source_file |                |          |

| IN                     | ds_t1w_mask_wf | OUT                  |
|------------------------|----------------|----------------------|
| inputnode.mask_file    |                | outputnode.mask_file |
| inputnode.source_files |                |                      |

| IN                     | register_template_wf | OUT                     |
|------------------------|----------------------|-------------------------|
| inputnode.lesion_mask  |                      | outputnode.anat2std_xfm |
| inputnode.moving_image |                      | outputnode.std2anat_xfm |
| inputnode.moving_mask  |                      | outputnode.template     |

| IN                 | anat_ribbon_wf | OUT                    |
|--------------------|----------------|------------------------|
| inputnode.pial     |                | outputnode.anat_ribbon |
| inputnode.ref_file |                |                        |
| inputnode.white    |                |                        |

| IN       | fast | OUT                  |
|----------|------|----------------------|
| in_files | fsl  | partial volume files |
|          |      | partial volume map   |

| IN                   | fsLR_reg_wf | OUT                        |
|----------------------|-------------|----------------------------|
| inputnode.sphere_reg |             | outputnode.sphere_reg_fsLR |

| IN                     | cortex_masks_wf | OUT                     |
|------------------------|-----------------|-------------------------|
| inputnode.midthickness |                 | outputnode.cortex_masks |
| inputnode.thickness    |                 | outputnode.source_files |

| IN                     | ds_ribbon_mask_wf | OUT                  |
|------------------------|-------------------|----------------------|
| inputnode.mask_file    |                   | outputnode.mask_file |
| inputnode.source_files |                   |                      |

| IN                     | ds_template_registration_wf | OUT                     |
|------------------------|-----------------------------|-------------------------|
| inputnode.anat2std_xfm |                             | outputnode.anat2std_xfm |
| inputnode.source_files |                             | outputnode.std2anat_xfm |
| inputnode.std2anat_xfm |                             | outputnode.std2anat_xfm |
| inputnode.template     |                             |                         |

| IN      | lut_t1w_dseg | OUT |
|---------|--------------|-----|
| in_dseg | utility      | out |

| IN     | fast2bids | OUT |
|--------|-----------|-----|
| inlist | utility   | out |

| IN                        | ds_fsLR_reg_wf | OUT                        |
|---------------------------|----------------|----------------------------|
| inputnode.source_files    |                | outputnode.sphere_reg_fsLR |
| inputnode.sphere_reg_fsLR |                |                            |

| IN                     | ds_cortex_masks_wf | OUT                   |
|------------------------|--------------------|-----------------------|
| inputnode.mask_files   |                    | outputnode.mask_files |
| inputnode.source_files |                    |                       |

| IN                     | ds_dseg_wf | OUT                  |
|------------------------|------------|----------------------|
| inputnode.anat_dseg    |            | outputnode.anat_dseg |
| inputnode.source_files |            |                      |

| IN                     | ds_tpms_wf | OUT                  |
|------------------------|------------|----------------------|
| inputnode.anat_tpms    |            | outputnode.anat_tpms |
| inputnode.source_files |            |                      |

| IN              | fsLR_buffer | OUT             |
|-----------------|-------------|-----------------|
| sphere_reg_fsLR | utility     | sphere_reg_fsLR |

| IN  | anat2std_buffer | OUT |
|-----|-----------------|-----|
| in2 | utility         | out |

| IN                        | msm_sulc_wf | OUT                        |
|---------------------------|-------------|----------------------------|
| inputnode.sphere          |             | outputnode.sphere_reg_fsLR |
| inputnode.sphere_reg_fsLR |             |                            |
| inputnode.sulc            |             |                            |

| IN       | seg_buffer | OUT      |
|----------|------------|----------|
| t1w_dseg | utility    | t1w_dseg |
| t1w_tpms |            | t1w_tpms |

| IN                       | ds_msm_sulc_wf | OUT                       |
|--------------------------|----------------|---------------------------|
| inputnode.source_files   |                | outputnode.sphere_reg_msm |
| inputnode.sphere_reg_msm |                |                           |

| IN  | template_buffer | OUT |
|-----|-----------------|-----|
| in2 | utility         | out |

| IN             | msm_buffer | OUT            |
|----------------|------------|----------------|
| sphere_reg_msm | utility    | sphere_reg_msm |

| IN  | std2anat_buffer | OUT |
|-----|-----------------|-----|
| in2 | utility         | out |

| IN               | outputnode | OUT            |
|------------------|------------|----------------|
| anat2std_xfm     |            | anat2std_xfm   |
| anat_ribbon      |            |                |
| cortex_mask      |            | subject_id     |
| fsnative2t1w_xfm |            |                |
| midthickness     |            |                |
| pial             |            |                |
| sphere           |            | subjects_dir   |
| sphere_reg       |            |                |
| sphere_reg_fsLR  |            |                |
| sphere_reg_msm   | utility    | t1w_dseg       |
| std2anat_xfm     |            |                |
| subject_id       |            |                |
| subjects_dir     |            | t1w_mask       |
| sulc             |            |                |
| t1w_dseg         |            |                |
| t1w_mask         |            | t1w_preproc    |
| t1w_preproc      |            |                |
| t1w_tpms         |            |                |
| t1w_valid_list   |            | t1w_valid_list |
| t2w_preproc      |            |                |
| template         |            | template       |
| thickness        |            |                |
| white            |            |                |

| IN                           | anat_reports_wf | OUT |
|------------------------------|-----------------|-----|
| inputnode.anat2std_xfm       |                 |     |
| inputnode.source_file        |                 |     |
| inputnode.subject_id         |                 |     |
| inputnode.subjects_dir       |                 |     |
| inputnode.subjects_dir       |                 |     |
| inputnode.t1w_conform_report |                 |     |
| inputnode.t1w_dseg           |                 |     |
| inputnode.t1w_mask           |                 |     |
| inputnode.t1w_preproc        |                 |     |
| inputnode.template           |                 |     |